

Chapter 10

1. Recall $\det(AB) = \det(A) \det(B)$

Thus $\phi(AB) = \phi(A) \phi(B)$

 $\therefore$ The mapping is an homomorphism3. Let $ax+b, cx+d \in \mathbb{R}[x]$

$$\phi[(ax+b) + (cx+d)] = \phi[x(a+c) + b+d]$$

$$= a+c = \phi(ax+b) + \phi(cx+d)$$

 $\therefore$ The mapping is an homomorphism from $\mathbb{R}[x]$ to $\mathbb{R}$ 6. Let $ax+b, cx+d \in G$

Recall $\int f(x) + g(x) = \int f(x) + \int g(x)$

Thus $\phi[(ax+b) + (cx+d)] = \phi(ax+b) + \phi(cx+d) + K$

Given $\int(ax+b) + \int(cx+d)$ passes through $(0,0)$,
find K such that $(\int(ax+b) + \int(cx+d) + K)|_{x=0} = 0$ Given that $\int(ax+b) \neq \int(cx+d)$ both pass through $(0,0)$

$$\int(ax+b)|_{x=0} = \int(cx+d)|_{x=0} = 0$$

Thus $0 + 0 + K = 0 \Rightarrow K = 0$

 $\therefore \phi$ is a group homomorphismb. $\ker \phi = \{0\}$

c. No.

7. Let $a, b \in G$

$$\sigma \phi(ab) = \sigma[\phi(ab)]$$

Since ϕ is a homomorphism

$$\sigma[\phi(ab)] = \sigma[\phi(a) \phi(b)]$$

Since σ is a group homomorphism

$$\sigma[\phi(a) \phi(b)] = (\sigma \phi)(a) \cdot (\sigma \phi)(b)$$

 $\therefore \sigma \phi$ is an homomorphism from G to K

9. Define $\phi: (g, h) \rightarrow g$ from $G \oplus H$ to G
 $\phi[(g_1, h_1)(g_2, h_2)] = \phi[(g_1 g_2, h_1 h_2)] = g_1 g_2$
 $= \phi[(g_1, h_1)] \phi[(g_2, h_2)]$
 $\text{Ker } \phi = \{(e, h) \mid h \in H\}$

14. Define $\phi: X \rightarrow 3X$ from $\mathbb{Z}_{12}$ to $\mathbb{Z}_{10}$
 $\phi(x) = 3x \bmod 10$
 Let $H = \{0, 6\}$. Observe $H \leq G$
 $\phi(H) = \{0, 8\} \not\subseteq \mathbb{Z}_{10}$ since $8+8 \bmod 10 = 6 \notin \phi(H)$
 $\therefore \phi$ is not a homomorphism

15. Let ϕ be a homomorphism from $\mathbb{Z}_{50}$ to $\mathbb{Z}_{50}$,
 $\text{Ker } \phi = \{0, 10, 20\}$, $\phi(23) = 9$
 $\phi^{-1}(9) = 23 + \text{Ker } \phi$
 $= \{23, 3, 13\}$

19. Recall $\text{Ker } \phi \leq \mathbb{Z}_{17}$
 Thus $|\text{Ker } \phi| \mid |\mathbb{Z}_{17}|$
 $|\text{Ker } \phi| = \{1, 17\}$ and $|\text{Ker } \phi| \neq 1$
 $\therefore \phi$ maps all elements in $\mathbb{Z}_{17}$ to $\bar{e}$

22. Define $\phi: G \rightarrow \bar{G}$ by $\phi(g) = \bar{g}$
 Let $|g| = 8$ [show $\exists g \in G$ with order 8]
 Let $|g| = m$, $\phi(g) = \bar{g}$
 $\phi(g^m) = [\phi(g)]^m = \bar{g}^m = \bar{e}$
 Since $\bar{g}$ has order 8, $\bar{g}^m = \bar{e}$ implies $8 \mid m$
 Thus $m = 8z$ for $z \in \mathbb{Z}$
 Consider element $g^z \in G$
 $|g^z| = \frac{m}{\gcd(m, z)} = \frac{8z}{\gcd(8z, z)} = \frac{8z}{z} = 8$
 $\therefore g^z \in G$ with order 8

25. Define mapping $\phi: \mathbb{Z}_{20} \rightarrow \mathbb{Z}_{10}$

$$\gcd(20, 10) = 10$$

$\therefore$ There are 10 homomorphisms from $\mathbb{Z}_{20}$ to $\mathbb{Z}_{10}$

b. Let ϕ be an auto homomorphism

To map generator to generator, find $\phi(10)$

[Euler-phi function]

$\phi(10) = 4$. This means there are 4 elements in $\mathbb{Z}_{10}$ with order 10

$\therefore$ There are 4 homomorphisms from $\mathbb{Z}_{20}$ into $\mathbb{Z}_{10}$

30. Define mapping $\phi: G \rightarrow \mathbb{Z}_6 \oplus \mathbb{Z}_2$ by G onto $\mathbb{Z}_6 \oplus \mathbb{Z}_2$

$$\text{Let } |\text{Ker } \phi| = 5$$

$$\text{We know } G / \text{Ker } \phi \cong \mathbb{Z}_6 \oplus \mathbb{Z}_2$$

$$\frac{|G|}{5} = 6 \cdot 2$$

$$|G| = 60$$

All subgroups of $\mathbb{Z}_6 \oplus \mathbb{Z}_2$ are normal because $\mathbb{Z}_6 \oplus \mathbb{Z}_2$ is Abelian

Pulling it back to group G , for all $H \leq \mathbb{Z}_6 \oplus \mathbb{Z}_2$
 $\phi^{-1}(H) = |\text{Ker } \phi| \cdot |H|$

Possible orders for H : 1, 2, 3, 4, 6, 12

Possible orders for $\phi^{-1}(H)$: 5, 10, 15, 20, 30, 60

Since $H \leq \mathbb{Z}_6 \oplus \mathbb{Z}_2$, $\phi^{-1}(H) \leq G$

84. Suppose for contradiction there exists homomorphism $\phi: A_4 \rightarrow \mathbb{Z}_2$

This means $A_4 / \text{Ker } \phi \cong \mathbb{Z}_2$

$$|\text{Ker } \phi| = |A_4| \div |\mathbb{Z}_2| = 12/2 = 6$$

However A_4 has no normal subgroup of order 6

$\therefore$ There no onto homomorphism here as that implies $|C_{A_4}(c_i)|$ equal 2 which isn't true

36. Define mapping: $\phi: \mathbb{Z} \oplus \mathbb{Z} \rightarrow G$

$$(4, 4) = x(3, 2) + y(2, 1)$$

$$4 = 3x + 2y$$

$$4 = 2x + y$$

$$x = 4, y = -4$$

$$\phi[(4, 4)] = 4\phi(3, 2) - 4\phi(2, 1) = 4a - 4b$$

41. Let G be a group, $K \leq G$, $N \triangleleft G$

$KN = \{kn \mid k \in K, n \in N\}$. Since $N \triangleleft G$, $KN \leq G$

Define mapping: $\phi: K \rightarrow KN/N$ by $\phi(k) = kN$

$$\text{Let } k_1, k_2 \in K. \quad \phi(k_1 k_2) = k_1 k_2 N = k_1 N k_2 N = \phi(k_1) \phi(k_2)$$

Thus ϕ is a homomorphism

$$\text{Ker } \phi = \{k \in K \mid k \in N\}$$

$$\text{Hence } \text{Ker } \phi = K \cap N$$

$\therefore$ By First isomorphism theorem,
 $K / K \cap N \cong KN / N$

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In a cyclic group, there exists unique subgroups with some unique order.

Thus finding all generators will provide all possible mapping because $\phi(g^n) = [\phi(g)]^n \quad \forall g \in G$

55. Define mapping $\phi: \mathbb{Z}[x] \rightarrow \mathbb{Z}$ by $\phi(f) = f(3)$

$$\phi(f+g) = (f+g)(3) = f(3) + g(3) = \phi(f) + \phi(g)$$

Thus ϕ is a homomorphism from $\mathbb{Z}[x]$ to $\mathbb{Z}$

$\text{Ker } \phi$ is set of all polynomial in $\mathbb{Z}[x]$ that has solution $y=0$ at $x=3$